

Kevin Ndayishimiye

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EDUCATION

Howard University

Bachelor of Science in Mathematics, Concentration in Applied Mathematics

– Minor: Computer Science

– Cumulative GPA: 3.8/4.0

Washington, D.C.

Aug 2024 – May 2028

RESEARCH EXPERIENCE

Undergraduate Researcher (Geometric Optics)

Howard University – Supervisor: Dr. Henok Mawi

Dec 2025 – Present

Washington, D.C.

- Investigating the **Prescribed Intensity Problem**, a non-linear inverse problem focused on designing freeform lens geometries that map source light distributions to target measures.
- Bridging classical optics with **Optimal Transport theory**, utilizing the **Monge-Ampère Equation** to model the transformation of probability densities via refraction and reflection.
- Developed a computational framework in **MATLAB** for simulating ray-tracing on non-standard surfaces; currently transitioning to **numerical solvers** for non-linear Partial Differential Equations (PDEs).

ACADEMIC PROJECTS

Vote Estimator (Ecological Inference) | Python, PyQGIS, NumPy

Dec 2025 – Present

- Developing a custom **Ecological Inference (EI) model** to estimate latent vote probability vectors ($P \in \mathbb{R}^{i \times d \times k}$) from aggregate precinct data.
- Formulated a compound loss function combining **L2 data-fit**, **neighbor-weighted Laplacian smoothing**, and **entropy regularization** to solve for non-degenerate probability distributions.
- Implementing **convex optimization** via the Adam optimizer to minimize loss over the probability simplex, utilizing PyQGIS to construct spatial adjacency matrices.

SpadesAI (Reinforcement Learning) | Python, TensorFlow, C++

July 2025 – Present

- Engineered a game-playing agent using **Monte Carlo Tree Search (MCTS)** guided by deep neural networks, similar to the AlphaZero architecture.
- Implemented **PUCT (Predictor Upper Confidence Trees)** to balance exploration/exploitation across a state space of $\approx 10^{20}$.
- Trained three distinct neural networks (Bidding, Playing, and Value) via **self-play reinforcement learning**, achieving a 2100+ Elo rating against human players.

LEADERSHIP & COMMUNITY SERVICE

Youth Coordinator

East Side Parkways Coalition

Oct 2023 – Present

Buffalo, NY

- Organized community canvassing efforts to support the restoration of Humboldt Parkway.
- Assisted in fundraising efforts supporting legal action against the NYSDOT.
- Contributed to a campaign that successfully resulted in a court injunction requiring an **Environmental Impact Statement (EIS)**.

TECHNICAL SKILLS

Mathematical Computing: MATLAB, Python (NumPy/SciPy), R, LaTeX

Statistical Modeling: Ecological Inference, Markov Random Fields, Monte Carlo Methods, Reinforcement Learning

Numerical Methods: Non-linear Optimization (Adam, Gradient Descent), Numerical PDEs, Ray Tracing